

GENG0035 Web Development: Library Management System

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System Overview:

Name: Library Management System

Purpose: To offer librarians digital assistance in managing books, members and borrowings from the library.

Description:

This web application provides the necessary tools for improved accessibility and efficiency in managing different aspects of a librarian's work. Firstly, providing basic statistics of the library's activities including any recent borrowings, number of books and members as well as member activity. Next, the website also features tools to help manage library books through adding, editing or deleting books from the library database. Additionally, tools for managing library members are also present in a similar sense, where new members can be added or details of pre-existing members can be adjusted. Finally, establishing a book borrowing system where users assign books to members under a set timeframe as borrowed books.

Core features:

- Secure Authentication (Login/Register/Logout)
- Dashboard with library statistics and recent activity. Also features search bar navigation for books and members.
- Book management where books can be added, edited, viewed and deleted from the library system. Books can also be searched and filtered by keywords and categories using the search bar.
- Member management where members can be added, edited, viewed and deleted from the library system. Members can also be searched by Member or ID from the search bar.
- Book borrowings management for assigning books to be borrowed and to whom, as well as confirmation of return and notification overdue books.

System architecture:

library_management

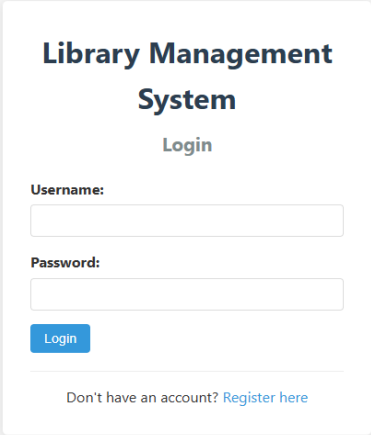
- **assets**
 - CSS
 - style.css
 - js
 - script.js
- **auth**
 - login.php
 - logout.php
 - register.php
- **books**
 - add.php
 - delete.php
 - edit.php
 - index.php
- **members**
 - add.php
 - delete.php
 - edit.php
 - index.php
- **borrow**
 - borrow.php
 - index.php
- **includes**
 - db_connect.php
 - header.php
 - footer.php
- index.php
- library_management.sql
- READ.me
- video

Technologies used:

- HTML, CSS and JavaScript for front-end development
- PHP for back-end and API management
- MySQL for database and data management.
- XAMPP Apache server and my PHP admin.
- No frameworks used.
- W3schools website for reference material

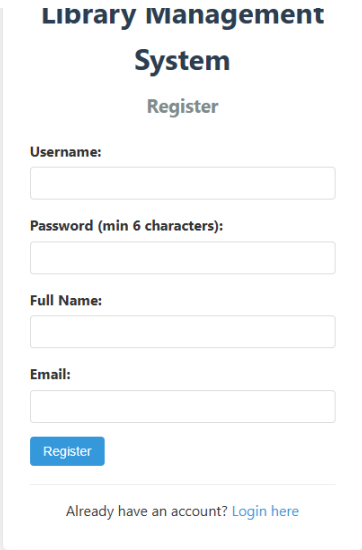
USER MANUAL:

- 1) First, you would be redirected to the authentication page. Here, you would need to login to your user account by entering your username and password as well as click on the Login button in order to proceed to the home page. If you have not made an account, then you could click on the **“Register here”** text to create one.



The image shows a login form for the Library Management System. The form is centered on a light gray background. It has a title 'Library Management System' in bold, followed by the word 'Login' in a smaller font. Below the title are two input fields: 'Username:' and 'Password:'. A blue 'Login' button is positioned below the password field. At the bottom of the form, there is a link that says 'Don't have an account? [Register here](#)'.

- 2) In the register page, you would need to enter a username, a secure password, your full name and your email address. Then, simply click on the Register button to create an account. You will be redirected back to the Login Page where you can login. You may also click on the **“Login here”** text if you already have an account.



The image shows a register form for the Library Management System. The form is centered on a light gray background. It has a title 'Library Management System' in bold, followed by the word 'Register' in a smaller font. Below the title are four input fields: 'Username:', 'Password (min 6 characters):', 'Full Name:', and 'Email:'. A blue 'Register' button is positioned below the email field. At the bottom of the form, there is a link that says 'Already have an account? [Login here](#)'.

- 3) After authentication, you will finally be on the home page where you are first greeted by the Dashboard. Here, simple statistics such as Total Books, Total Members, Books Borrowed and Overdue Books are displayed alongside any recent activity of borrowings from members. On the top right, there is also a search bar that allows searching for books using any keyword. On the side bar, you can access the different functionalities.

Library Management
Welcome, Ahmad Abbas Zahin

Dashboard

Search books...

Total Books
5

Total Members
1

Books Borrowed
0

Overdue Books
0

Recent Activity

Book	Member	Borrowed Date	Due Date	Status
Power and Prediction: The Disruptive Economics of Artificial Intelligence	Ahmad Yusuf Zahin Ahmad Azrin	2025-07-31	2025-08-14	Returned
naked economics: Undressing The Dismal Science	Ahmad Yusuf Zahin Ahmad Azrin	2025-07-30	2025-08-13	Returned
THE HOBBIT	Ahmad Yusuf Zahin Ahmad Azrin	2025-07-30	2025-08-13	Returned

Accessing Book Management:

Displays all books in a table format with their specifications and features the option to **edit** or **delete** books under the actions tab on the far right of the table. Includes a Search bar which has the added benefit of filtering by category or book specification. Option to add a new book with the button on the top right.

Library Management
Welcome, Ahmad Abbas Zahin

Books Management

Search by title, author or ISBN All Categories

Title	Author	ISBN	Category	Year	Available	Actions
naked economics: Undressing The Dismal Science	charles wheelan	978-0-393-35649-6	Biography	2002	1	Edit Delete
Power and Prediction: The Disruptive Economics of Artificial Intelligence	Ajay Agrawal, Joshua Gans, Avi Goldfarb	978-1-66-48824-198	Technology	2023	1	Edit Delete
THE CAVES OF STEEL	Isaac Asimov, Mike Topping	978-0-00-827776-5	Science Fiction	2018	3	Edit Delete

Navigating the “Add Book” Page:

Once the user has entered into the page, they would be given 7 input fields to fill out for the book they are adding to the database. “Title” for the name of the book, “Author” for the writer as well as supporting writers and illustrators, “ISBN” for the isbn code of the book, “Category” for the genre of the book, “Publication” for the year the book was published, “Copies Available” for the quantity of the books available for borrowing and “Description” for a general synopsis of the book. Finally, the user just has to click on the “Add book” button to add the book.

Library Management

Welcome, Ahmad Abbas Zahin

Dashboard

Books Management

Members Management

Borrow/Return

Logout

Add New Book

Back to Books

Title:

Author:

ISBN:

Category:

-- Select Category --

Publication Year:

Copies Available:

1

Description:

Add Book

Accessing Member management:

Displays all the users in a table format with their credentials of ID, Name, Emails, Phone number, membership date and their library status. Also features the option to Edit or Delete users from the database under the actions tab on the far right of the table. Includes a search bar for users to search for members by name or ID and the Option to Add a New Member to the database above the search bar.

Library Management

Welcome, Ahmad Abbas Zahin

Dashboard

Books Management

Members Management

Borrow/Return

Logout

Members Management

Add New Member

Search by name or member ID Search

ID	Name	Email	Phone	Membership Date	Status	Actions
1	Ahmad Yusuf Zahin Ahmad Azrin	yusuf.zahin@gmail.com	0162107567	2025-07-30	Active	Edit Delete

Navigating the “Add Member” Page:

Once the user has entered into the page, they would be given 7 input fields to fill out for the member they are adding to the database. “First Name” and “Last Name” for the full name of the member, “Email” for the email address of the member, “Phone” for member phone number, “Address” for the user’s address, “Membership date” for when the member was registered and “Status” for member’s library activity. Finally, The user just has to click on the “Add member” to add the new member.

Library Management

Welcome, Ahmad Abbas Zahin

Dashboard

Books Management

Members Management

Borrow/Return

Logout

Add New Member

Back to Members

First Name:

Last Name:

Email:

Phone:

Address:

Membership Date:

04 / 08 / 2025

Status:

Active

Add Member

Accessing Borrowings Management:

Displays all previous and currently borrowed books in a table format consisting of the book borrowed, the member borrowing the book, the date the book was borrowed, the date the book has to be returned and the status of the borrowing. There is also the option to Return the books if the member had returned the book to the library under the actions tab on the far right of the table. This page also includes the option to borrow a book as a button on the top right which directs the user to another page.

Library Management

Welcome, Ahmad Abbas Zahin

Dashboard

Books Management

Members Management

Borrow/Return

Logout

Borrow/Return Books

Borrow a Book

Book borrowed successfully! Due date: August 22, 2025

Book	Member	Borrowed Date	Due Date	Status	Actions
naked economics: Undressing The Dismal Science	Ahmad Yusuf Zahin Ahmad Azrin	2025-08-08	2025-08-22	Borrowed	Return
naked economics: Undressing The Dismal Science	Ahmad Elyas Zahin Ahmad Azrin	2025-08-08	2025-08-22	Borrowed	Return
THE HOBBIT	Ahmad Yusuf Zahin Ahmad Azrin	2025-07-30	2025-08-13	Returned	
naked economics: Undressing The Dismal Science	Ahmad Yusuf Zahin Ahmad Azrin	2025-07-30	2025-08-13	Returned	
Power and Prediction: The Disruptive Economics of	Ahmad Yusuf Zahin	2025-07-30	2025-08-13	Returned	

Navigating the “Book Assignment” Page:

Once the user has entered into the page, they would be given 2 input fields to fill out for the book borrowing by a member. First being “Book” for the book being borrowed and then “Member” for the member borrowing the book. Additionally, the user can also observe the date when the book was borrowed and when it should be returned. Finally, The user just has to click on the “Borrow Book” button to assign the book to the member for borrowing.

Library Management

Welcome, Ahmad Abbas Zahin

Dashboard

Books Management

Members Management

Borrow a Book

Logout

Borrow a Book

[Back to Borrowings](#)

Book:

-- Select Book --

Member:

-- Select Member --

Borrowed Date:
August 4, 2025

Due Date:
August 18, 2025

[Borrow Book](#)